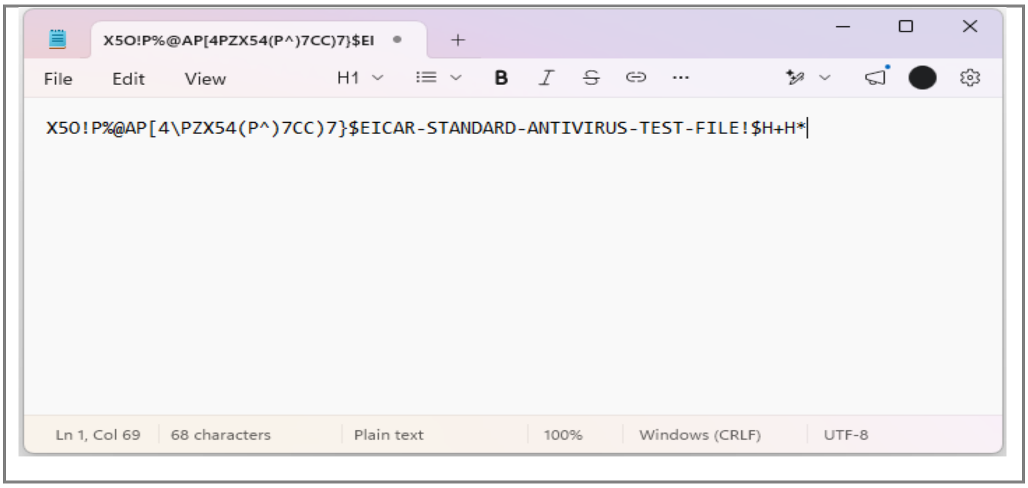
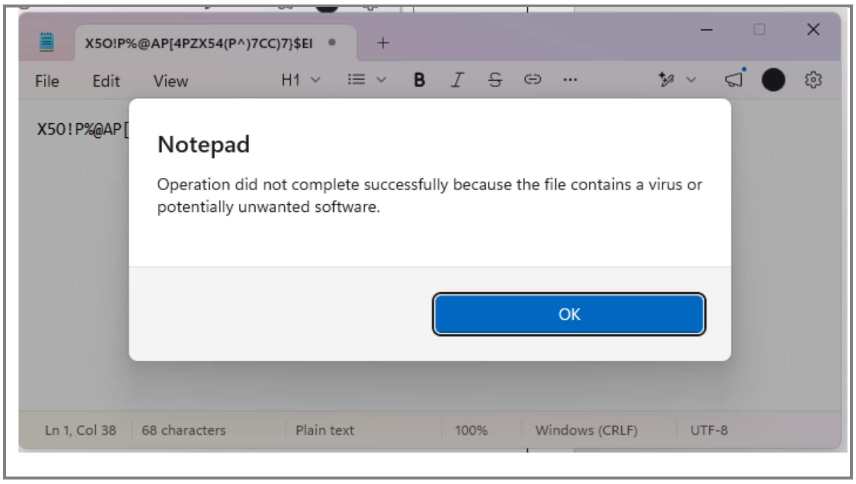
AIM: To understand the basic structure of a malware sample by creating the harmless EICAR antivirus test file and a simple Python file-copier script, and to observe how Windows Defender detects the test sample.

PROCEDURE:

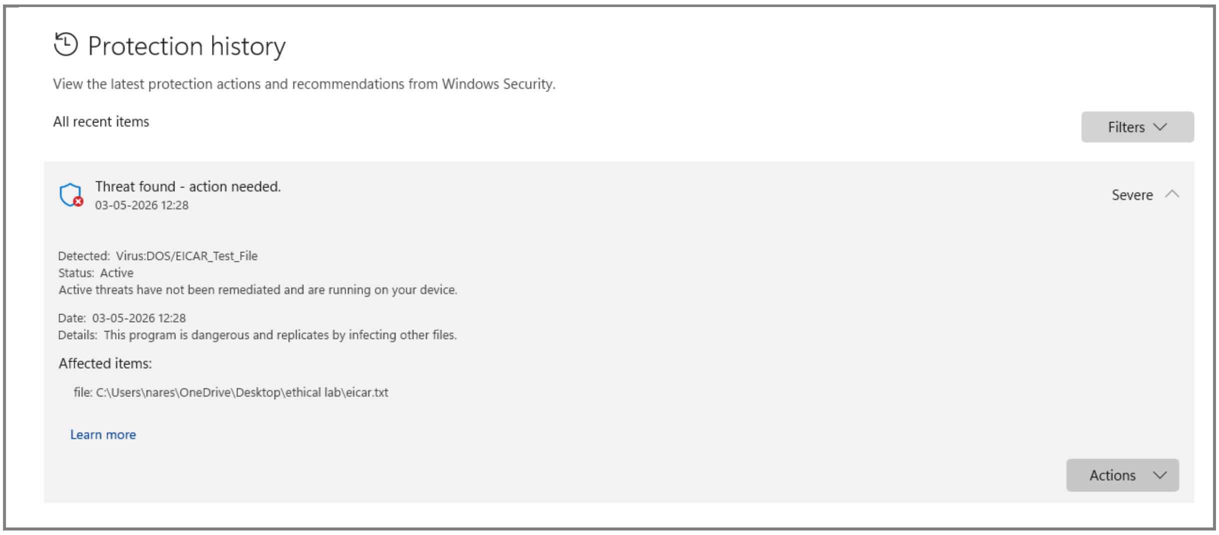
1. Open Notepad and paste the standard EICAR test string (X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*) and save it on the Desktop as eicar.txt.



2. Save the file. Windows Defender immediately quarantines it and shows a 'Threat found' notification, proving that the sample is recognised as a virus signature.

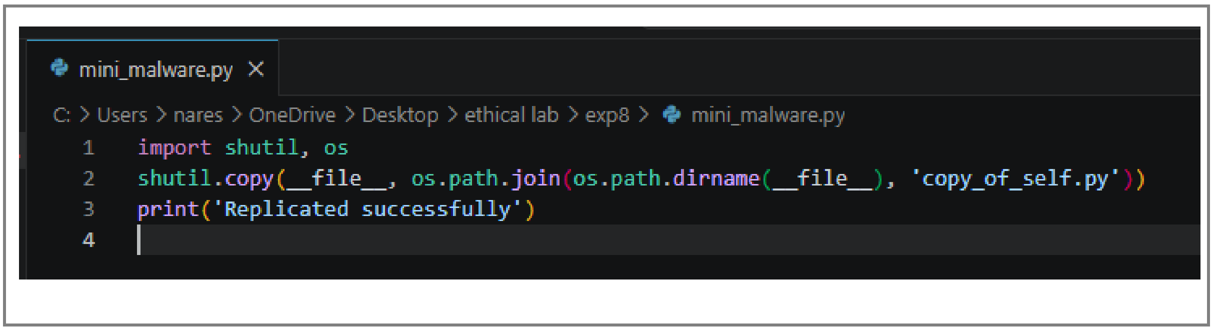


3. Open Windows Security > Virus & threat protection > Protection history and verify that EICAR_Test_File has been blocked.

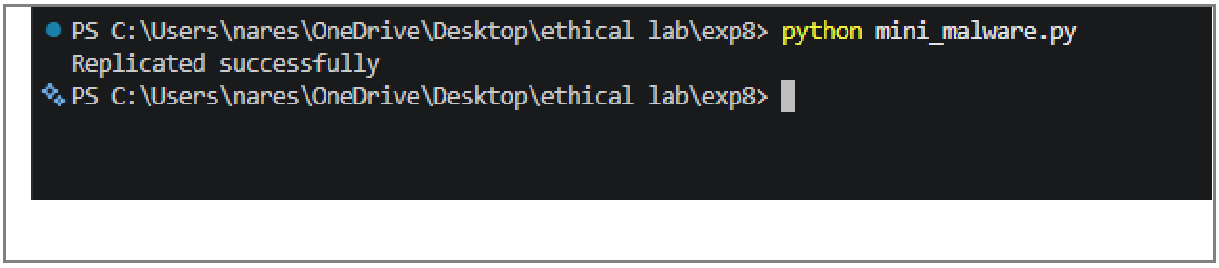


4. Create a new Python file mini_malware.py containing a simple self-replicating script that copies itself into the same folder with a new name (educational only):

```
import shutil, os
shutil.copy(__file__, os.path.join(os.path.dirname(__file__), 'copy_of_self.py'))
print('Replicated successfully')
```



5. Open Command Prompt, navigate to the file location and run 'python mini_malware.py'. Observe that a duplicate file 'copy_of_self.py' is created in the same folder.



6. Delete both files after the demonstration. Conclude that even a very small program can exhibit malware-like replication behaviour, and that signature-based AV tools such as Defender successfully detect known samples.

RESULT: Thus, the basic concept of malware was understood by generating the EICAR test sample and a simple Python replicator, and detection by Windows Defender was demonstrated successfully.